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B-3

1 $x_t = 0.5$
$h_{t-1} = 0.2$
$c_{t-1} = 0.4$

LSTM eqns are:-

$$f_t = \sigma(w_f x_t + v_f h_{t-1} + b_f)$$
$$i_t = \sigma(w_i x_t + v_i h_{t-1} + b_i)$$
$$\tilde{C}_t = \tanh(w_c \cdot x_t + v_c \cdot h_{t-1} + b_c)$$
$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t$$
$$o_t = \sigma(w_o \cdot x_t + v_o h_{t-1} + b_o)$$
$$h_t = o_t \cdot \tanh(c_t)$$

a) forget gate:-

$$z_f = (0.7)(0.5) + (0.3)(0.2) + 0.1$$
$$= 0.5100$$

$$f_t = \sigma(z_f)$$
$$= \frac{1}{1+e^{-0.5100}} = 0.6248$$

b) Input gate

$$z_i = (0.6)(0.5) + (0.4)(0.2) + 0$$
$$= 0.3800$$

$$i_t = \frac{1}{1+e^{-0.3800}} = 0.5939$$

c) Candidate cell state ($\tilde{c}_t$)

$$z_c = (0.9)(0.5) + (0.2)(0.2)$$
$$= 0.4900$$

$$\tilde{c}_t = \tanh(0.4900)$$
$$= 0.4542$$

d) new cell state (c_t):

$$c_t = f_t \cdot c_{t-1} + i_t \cdot \tilde{c}_t$$
$$c_t = (0.6248)(0.4) + (0.5939)(0.4542)$$
$$= 0.5197$$

e) Output gate (o_t):

$$z_o = (0.5)(0.5) + (0.5)(0.2) + 0.1$$
$$= 0.4500$$

$$o_t = \frac{1}{1+e^{-0.4500}} = 0.6106$$

f) New hidden state (h_t):

$$h_t = o_t \cdot \tanh(c_t)$$
$$\therefore \tanh(0.5197) \approx 0.4775$$
$$h_t = (0.6106)(0.4775)$$
$$\approx 0.2915$$

2 $d = 10$
$h = 32$

(a) Params $= 4(dh + h^2 + h)$
$= 4(10 \times 32 + 32 \times 32 + 32)$
$= 4(320 + 1024 + 32)$
$= 4(1376)$
$= 5504$

(b) Two bias vectors per gate (PyTorch)
Params $= 4(dh + h^2 + 2h)$
$= 4(320 + 1024 + 64)$
$= 4(1408)$
$= 5632$
difference $= 5632 - 5504 = 128$

3

(a) $f = 0.9$
$k = 10$
$= 0.9^{10} = 0.3487$

$K = 50$
$= 0.9^{50} = 0.0052$

(b) $K = 50$
$f = 0.5$
$= 0.5^{50} = 8.8818 \times 10^{-16}$

$f = 0.99$
$= 0.99^{50} = 0.6050$

c) Explanation

When f is small, the gradient decreases very quickly over many time steps, making it hard to learn long term dependencies. When f is close to 1, the gradient is preserved much better.

4

(a) Total hourly records

= 3 years × 365 days × 24 hours

= 1095 × 24

= 26,280

(b) Number of input/output windows

N = 26,280

L = 168

H = 24

$N - L - H + 1$

= 26280 − 168 − 24 + 1

= 26,089

(c) Tensor shapes

Input batch (64, 168, 3

Output batch (64, 24)

(d) Training windows (70%)

$N_{train} = 0.70 \times 26,280 = 18396$

$W_{train} = 18396 - 168 - 24 + 1$

$= 18205$

Mini-batches (batch size 64)

$= \text{ceil}(18205/64)$

$= 285$ (284 full + 1 partial)

(e) Params $= 4(dh + h^2 + h)$

$= 4[(3 \times 64) + (64)^2 + 64]$

$= 17408$

5

(a) Min-max normalisation

$$x_{norm} = \frac{x - x_{min}}{x_{max} - x_{min}}$$

$$\Rightarrow \frac{95 - 20}{120 - 20} = \frac{75}{100}$$

$= 0.7500$

(b) Prediction

Output probability $= 0.83$

Threshold $= 0.5$

Since $0.83 > 0.5$

$\Rightarrow$ predict failure

(c) Confusion matrix

Total = 1000
actual failures = 40
Predicted failures = 45 (30 correct)

TP = 30
FP = 45 - 30 = 15
FN = 40 - 30 = 10
TN = 1000 - (30 + 15 + 10) = 945

Metrics

$$\text{Precision} = \frac{TP}{TP+FP} = \frac{30}{45} = 0.6667$$

$$\text{Recall} = \frac{TP}{TP+FN} = \frac{30}{40} = 0.7500$$

$$F_1 = \frac{2PR}{P+R} = \frac{2(0.6667)(0.7500)}{0.6667 + 0.7500}$$

$$= 0.7059$$

$$\text{Accuracy} = \frac{TP+TN}{1000} = \frac{30+945}{1000}$$

$$= 0.9750$$

(d) Recall, because missing a real failure can lead to costly damage and downtime

Accuracy can be misleading here because dataset is highly imbalanced (only 40/1000 failures). A model that predicts no failures would still get 90% accuracy but detect zero failures.